

Outlook

The methodology developed in this thesis provides a foundation for accelerated vibration testing of complex assemblies. Several promising research directions can advance this approach and increase its accuracy, realism and usefulness for device level qualification.

Enhanced control through multiple feedback positions

This work relied on single point control at the PCB_LEFT position, which ensured stable regulation of the excitation but could not fully manage the spatial variation across a flexible printed circuit board. A natural improvement is the use of a second control position or a general multi-point control strategy. By regulating the excitation with feedback from multiple sensors, the test could reduce local deviations in fatigue damage and maximum response and create a more uniform accelerated environment. Furthermore, an SRS or FDS controlled shaker test could also be arranged.

Spatial mapping of fatigue and structure response

The current analysis used measurements at a limited number of sensor locations. A more complete view of structural behaviour can be achieved by constructing a spatial map of fatigue damage or maximum response across the entire printed circuit board. This can ideally be done using infinite sensor positions on the board but can be approximated through a combination of finite element modelling, modal information and measurements placed at strategically chosen points. Such a map would help identify the most critical regions and clarify how the accelerated profile distributes damage.

Adaptive accelerated profiles with resonance tracking

The accelerated profile used in this study remains fixed throughout the test, even though the resonances of the printed circuit board may shift as the structure ages. A more advanced approach would monitor the response during the test and adjust the spectral shape of the excitation as the dynamic properties change. This adaptive concept would keep the vibration signal aligned with the current behaviour of the assembly and provide a more realistic simulation of ageing and fatigue progression.

Strain-based durability analysis for field to laboratory comparison

Validation in this thesis was based on acceleration and derived spectral quantities. A stronger and more direct approach for future studies is to perform durability analysis through strain based measurements. Strain gauges placed on critical locations of the printed circuit board during real operation and at the same locations during the accelerated test would allow a direct comparison of the strain histories in field and laboratory conditions. This apples to apples comparison yields true durability equivalence because fatigue damage correlates more closely with strain than with acceleration. Such strain based durability analysis would therefore offer a decisive method for validating accelerated tests and confirming that the condensed profile reproduces the real structural loading.

Energy-based interpretation of fatigue

Fatigue can also be interpreted through the flow and dissipation of mechanical energy. Every vibration test inputs mechanical energy into the structure and this energy is divided among elastic storage, damping, radiation and irreversible damage. A long term research direction is to estimate the portion of energy that contributes to damage by tracking all major energy inputs and outputs. This perspective may provide deeper insight into the physics of degradation and lead to new ways of defining and validating accelerated tests.

Integration of thermal load in accelerated testing

The endurance environment of the backend PCB and subsequently Thermomix® TM7 is not purely mechanical. It also experiences thermal cycling, thermal gradients and temperature dependent material behaviour during operation. Future work may combine the vibration profile with a thermal acceleration component that reflects the temperature variations present in real service. This can involve alternating or simultaneous application of thermal loading and vibration to capture thermo mechanical fatigue interactions. Incorporating a thermal acceleration model would allow the accelerated test to reflect the full endurance environment rather than only the vibration component and may improve the accuracy of lifetime prediction for assemblies that are sensitive to temperature.